

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I **MD Imthiaz I(192521360)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

I.IMTHIAZ

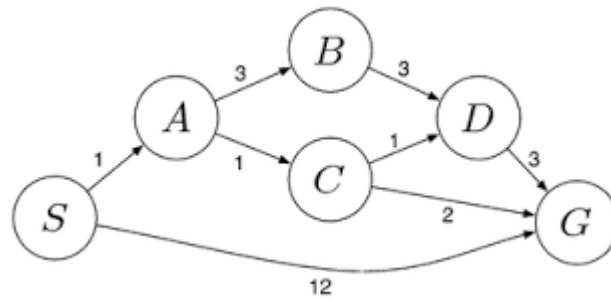
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



ANSWERS

1. Water Jug Problem:

Initial State:

- 4-gallon jug = 0
- 3-gallon jug = 0

Step	4-Gallon Jug	3-Gallon Jug	Action
1	0	3	Fill 3-gallon jug
2	3	0	Pour 3-gallon into 4-gallon
3	3	3	Fill 3-gallon again
4	4	2	Pour from 3-gallon to 4-gallon until 4-gallon is full
5	0	2	Empty the 4-gallon jug
6	2	0	Pour the remaining 2 gallons into the 4-gallon jug

Final State:

- 4-gallon jug = 2 gallons
- 3-gallon jug = 0 gallons

State Space Representation:

(0,0)

↓

(0,3)

↓

(3,0)



(3,3)



(4,2)



(0,2)



(2,0)

Conclusion:

Thus, exactly **2 gallons** of water are obtained in the **4-gallon jug**.

2. Mars Rover Agent

i. What are the percepts for this agent?

Percepts are the inputs received by the Mars Rover.

- Images from cameras
- Soil composition data
- Temperature readings
- Atmospheric pressure
- Terrain information
- Obstacle detection data
- Chemical analysis results
- GPS/Position information

ii. Characterize the Operating Environment

Property	Description
Observable	Partially Observable
Deterministic	Stochastic
Episodic/Sequential	Sequential
Static/Dynamic	Dynamic
Discrete/Continuous	Continuous
Single/Multi-agent	Single Agent

iii. Actions the Agent Can Take

- Move forward/backward
- Turn left/right
- Collect rock samples
- Drill the surface
- Capture images

- Analyze samples
- Send data to Earth
- Avoid obstacles
- Stop and recharge

iv. Performance Evaluation

The Mars Rover's performance can be evaluated using:

- Number of samples collected
- Accuracy of analysis
- Distance covered
- Energy efficiency
- Mission completion time
- Number of obstacles avoided
- Successful transmission of data

v. Suitable Agent Architecture

Learning Agent Architecture

Reason:

- Learns from previous experiences.
- Adapts to unknown environments.
- Improves navigation and obstacle avoidance.
- Makes intelligent decisions autonomously.

Conclusion

A **Learning Agent** is best suited because Mars is an uncertain and dynamic environment.

3. 8-Queens Problem

Problem Formulation

Initial State

- Empty chessboard.

State Space

- All possible placements of 8 queens.

Operators

- Place a queen in an empty square.

Goal State

- Place 8 queens so that no two queens attack each other.

Path Cost

- Each move costs 1.

Constraints

Queens must not share:

1. Same row
2. Same column
3. Same diagonal

Search Strategies

1. Breadth First Search (BFS)
2. Depth First Search (DFS)
3. Backtracking
4. Hill Climbing
5. Genetic Algorithm

State Space Search

Start



Place Queen 1



Place Queen 2



Check Constraints



Valid?



Yes → Continue

No → Backtrack



Repeat until 8 Queens are placed.

Conclusion

The 8-Queens problem is commonly solved using Backtracking Search, which efficiently explores the search space and eliminates invalid states.

4. OLA Cab Booking Application

Type of Problem Solving Agent

The most suitable agent is:

Goal-Based Agent

Reason

- Goal: Reach destination.
- Evaluates multiple cab options.
- Chooses the best route and cab type.

- Achieves the desired destination efficiently.

Pseudocode

BEGIN

INPUT source location

INPUT destination

DISPLAY available cab types
(Mini, Micro, Sedan, Prime, Shared)

USER selects cab type

CHECK availability

IF cab available THEN
 CALCULATE fare
 DISPLAY fare and arrival time

 IF user confirms booking THEN
 ASSIGN nearest driver
 SEND driver details
 START trip

 ELSE
 CANCEL request
 ENDIF

ELSE
 DISPLAY "No cabs available"
ENDIF

END

Working

1. User enters source and destination.
2. System shows available cabs.
3. User selects a cab.
4. System assigns the nearest driver.
5. Trip begins.

Conclusion

The OLA application acts as a **Goal-Based Problem Solving Agent** because its primary objective is to transport the customer from source to destination.

5. Uniform Cost Search (UCS)

Graph

$S \rightarrow A$ (1)

$S \rightarrow G$ (12)

$A \rightarrow B$ (3)

$A \rightarrow C$ (1)

$B \rightarrow D$ (3)

$C \rightarrow D$ (1)

$C \rightarrow G$ (2)

$D \rightarrow G$ (3)

Uniform Cost Search Execution

Path	Cost
$S \rightarrow G$	12
$S \rightarrow A$	1
$S \rightarrow A \rightarrow B$	4
$S \rightarrow A \rightarrow C$	2
$S \rightarrow A \rightarrow C \rightarrow D$	3
$S \rightarrow A \rightarrow C \rightarrow G$	4
$S \rightarrow A \rightarrow B \rightarrow D$	7
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6

Step-by-Step UCS

1. Start at **S (0)**.
2. Expand **A (1)**.
3. Expand **C (2)**.
4. Expand **D (3)**.
5. Expand **G (4)**.

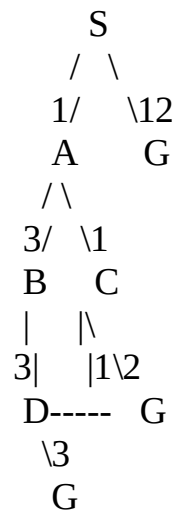
Least Cost Path

$S \rightarrow A \rightarrow C \rightarrow G$

Total Cost

$$1 + 1 + 2 = 4$$

Search Tree



Conclusion

- **Optimal Path:** $S \rightarrow A \rightarrow C \rightarrow G$
- **Minimum Cost:** 4

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem